

# Euler's Method

Section

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Given diffEq:  $\frac{dy}{dx} = x + y$

Let  $y = f(x)$  for particular sol.

$f(4) = -1$ , approximate  $f(4.2)$

with 2 steps of size 0.1

| x  | 4  | 4.1  | 4.2   |
|----|----|------|-------|
| y  | -1 | -0.7 | -0.36 |
| y' | 3  | 3.4  |       |

$$\textcircled{1} \quad y + 1 = 3(x - 4)$$

$$y + 1 = 3(4.1 - 4)$$

$$y + 1 = -0.3$$

$$\textcircled{2} \quad y + 0.7 = 3.4(x - 4.1)$$

$$f(4.2) = -0.36$$